# JAVASCRIPT

# □ PHASE 1 — JavaScript Foundations (Beginner)

### □ 1. Introduction to JavaScript

- What is JavaScript?
- Where JS is used (Frontend, Backend, AI, Automation)
- JS vs HTML/CSS
- How browser works (very simple)

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### □ 2. Setup & First Code

- Browser console
- VS Code setup
- Writing first JS code
- `script` tag (internal vs external)

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### □ 3. Variables & Data Types

- `var`, `let`, `const`
- Primitive vs non-primitive
- Strings, Numbers, Boolean, Null, Undefined

□ Mini Task:

- User intro program

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### □ 4. Operators & Conditions

- Arithmetic operators
- Comparison operators
- Logical operators

- `if`, `else`, `else if`, `switch`

 Project:

- Age checker (minor/adult)

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### 5. Loops (VERY IMPORTANT)

- `for`, `while`, `do while`
- `break`, `continue`

 Project:

- Multiplication table generator

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### 6. Functions (Core Concept)

- Function declaration
- Parameters & arguments
- Return keyword
- Arrow functions

 Project:

- Calculator

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# PHASE 2 — Core JavaScript (Strong Base)

### 7. Arrays & Objects

- Arrays (methods: push, pop, map, filter)
- Objects (real-world usage)

 Project:

- Student management system (basic)

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### □ 8. DOM Manipulation

- What is DOM?
- Selecting elements
- Changing content
- Events (click, input)

□ Project:

- Counter app
- Dark mode toggle

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### □ 9. Events & Event Handling

- Event listeners
- Forms handling

□ Project:

- Form validation

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### □ 10. Advanced Functions

- Callback functions
- Higher-order functions

□ Simple real-life examples

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# ⚡ PHASE 3 — Intermediate (Real Developer Level)

### □ 11. ES6+ Features (Must)

- Template literals
- Destructuring
- Spread & rest

- Default params

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### □ 12. Asynchronous JavaScript (□ Most Important)

- Callbacks
- Promises
- `async/await`

□ Project:

- API data fetch (Weather App)

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### □ 13. Error Handling

- `try...catch`
- Debugging tips

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### □ 14. Local Storage

□ Project:

- Todo App (with save data)

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# □ PHASE 4 — Advanced JavaScript

### □ 15. OOP in JavaScript

- Objects & classes
- Constructor
- Inheritance

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### □ 16. Closures & Scope (□ Interview Topic)

- Lexical scope
- Closure concept

 Explain with real-life analogy

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### 17. Hoisting & Execution Context

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### 18. This Keyword

 One of the most confusing topics

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### 19. Prototypes

 Deep JS understanding

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# PHASE 5 — Real World JavaScript

### 20. Working with APIs

 Real projects:

- Crypto price tracker
- GitHub profile finder

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### 21. Modules (Modern JS)

- import/export

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### 22. Clean Code & Best Practices

- Naming conventions
- Code structure

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### □ 23. Performance Optimization (Advanced)

- Debouncing
- Throttling

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# □ PHASE 6 — Projects

### □ Beginner Projects

- Calculator
- Todo App
- Quiz App

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### □ Intermediate Projects

- Weather App
- Notes App
- Expense Tracker

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### □ Advanced Projects

- Mini E-commerce
- Crypto Dashboard
- Chat App